

EXAMPLE 9**Greek Coupon Bond Implied Return**

Recall from Example 2 that seven-year Greek government bonds issued in 2020 with a 2.00 percent annual coupon had a price of EUR93.091 per EUR100 principal two years later.

1. What is the implied two-year return for an investor able to reinvest periodic interest at the original YTM of 2.00 percent?

Solution:

−1.445 percent

Using Equation 18 as for the discount bond, we must first calculate the future value (FV) after two years including the future price of 93.091 and all cash flows reinvested to that date:

$$FV_2 = PMT_1(1 + r) + PMT_2 + PV_2$$

$$FV_2 = 97.13 = 2 \times (1.02) + 2 + 93.091$$

$$r = -1.445\% = \left(\frac{97.13}{100}\right)^{\frac{1}{2}} - 1.$$

This negative annualized return was due to a EUR6.91 ($=100 - 93.091$) capital loss which exceeded the periodic interest plus reinvestment proceeds of EUR4.04 ($=2 \times (1.02) + 2$). The fall in the price of the bond to EUR93.091 in 2022 was a result of a rise in Eurozone inflation over the period.

For an investor purchasing the 2 percent coupon bond in 2022 at a price of EUR93.091, recall that we solved for an r (or YTM) of 3.532 percent. Note that this YTM calculation for the remaining five years assumes all cash flows can be reinvested at 3.532 percent through maturity. The rate(s) at which periodic interest can be reinvested is critical for the implied return calculation for coupon bonds as shown below. YTM is computed assuming that the bond is held to maturity.